C1M1: Gemini Updated

Course 1, Module 1: Foundations of the Innovation Mindset

Part 1: The Call - Setting the Stage & Chapter 1

By the end of this module, the learner will be able to:

1. **Define** the "innovation mindset" and **correlate** core clinical competencies (e.g. diagnostic reasoning, empathy) directly to established innovation frameworks like Design Thinking. *(Understand / Apply)*
2. **Analyze** the macro-economic and structural trends—including administrative burnout, healthcare consumerism, and technological acceleration—that necessitate intrapreneurial physician leadership. *(Analyze)*
3. **Differentiate** between a superficial "idea" and a rigorously vetted "opportunity" using the core tenets of Sandra Lam's *Intrapreneurship Formula*. *(Analyze / Evaluate)*
4. **Evaluate** systemic operational failures within their own clinical environment by conducting root-cause analyses and defending their findings in a peer-reviewed forum (the "M&M of Innovation"). *(Evaluate)*
5. **Create** a personal innovation baseline (SWOT analysis) and an "Opportunity Radar" that captures and categorizes real-world clinical frictions for future project development. *(Create)*

Slide 1: Welcome & The Modern Clinician's Dilemma

- **Visual Format:** Welcome Video (Talking Head + B-Roll) **Video Storyboard: • 0:00 - 0:15 (On-Camera):** You (Dr. Naik) are centered on screen in a professional, well-lit office or clinical setting. ◦ *On-Screen Text (Lower Third):* Tejal Naik, M.D. | Hospitalist & Entrepreneur-in-Residence **• 0:15 - 0:30 (B-Roll Overlay):** Footage of a busy hospital corridor blurred in the background, transitioning to a close-up of a physician looking tired while scrolling through a dense Electronic Health Record (EHR) system. **• 0:30 - 0:45 (On-Camera):** Return to you, speaking with a forward-leaning, encouraging posture. ◦ *On-Screen Text:* From Clinical Reactor → Systemic Innovator **• 0:45 - 1:00 (Animation Overlay):** A simple animation showing a traditional clinical pathway diverging into a new, streamlined "innovator-designed" pathway. **Prompt to Generate Video/B-Roll (for tools like Runway, Sora, or Synthesia):***Cinematic, 4k resolution video of a modern hospital environment. Scene 1: A physician looking thoughtfully at a computer screen, soft natural light,*

shallow depth of field. Scene 2: High-quality b-roll of a busy nursing station, fast-paced but professional. Color grading should be slightly cool, medical blues and clean whites, conveying urgency but professionalism.

- **Narration (Speaker Notes):** "Welcome. I am Dr. Tejal Naik. You are practicing in an era of immense technological advancement and increasing system complexity. You have been expertly trained to follow evidence-based pathways, but likely not taught how to create the pathways of the future. The primary barrier to change is rarely a lack of ideas; it is the absence of a structured process and the psychological safety to pursue them. Therefore, our first goal is to establish a new baseline. **An innovation mindset is the internal conviction that you are not just an employee within a healthcare system, but a designer of it, empowered to bridge the gap between current limitations and future possibilities.** The solution to healthcare's biggest challenges lies dormant within its most valuable asset: you."

- **Reference:** *National Academy of Medicine (NAM). (2019). Taking Action Against Clinician Burnout: A Systems Approach to Professional Well-Being. Washington, DC: The National Academies Press.* (Used to ground "The Modern Clinician's Dilemma" in systemic, peer-reviewed literature).

Slide 2: The Innovator's "Hero's Journey"

Visual Format: High-Quality Infographic / Roadmap Diagram

Prompt to Generate Image (for tools like Notebook LM or Midjourney or DALL-E 3):

A clean, modern corporate infographic of a horizontal winding roadmap titled 'The Innovator's Journey'. It features three distinct waypoints: 1. Sparking Innovation (Insight to Opportunity), 2. Building the Solution (Design to Pitch), 3. Driving Impact (Project to Change). The design uses a professional healthcare color palette of navy blue, teal, and white. Minimalist, flat vector icons for each stage. No text clutter, highly legible, optimized for a PowerPoint slide.

Narration (Speaker Notes): "This certificate is designed to unlock a skill set you already possess as clinicians. We will treat this program as a personal and professional quest—an adaptation of the classic Hero's Journey. Course 1, which you are starting now, is 'Sparking Innovation: From Insight to Opportunity,' where you will learn to see your daily clinical world differently. Course 2 is 'Building the Solution: Design, Plan, and Pitch'. Finally, Course 3 is 'Driving Impact: From Project to Lasting Change,' where you bring your vision to life. By the end of this journey, you won't just have new knowledge; you will have a tangible playbook and the foundation for an executive one-pager to make your ideas a reality."

Discussion Post 1: Self-Reflection & Bias Awareness (Ungraded)

- **Format:** Required asynchronous discussion board.
- **Prompt:** "Before we dive into the frameworks of innovation, let's establish where you are starting from. Please introduce yourself and your clinical or administrative role. Then, reflect on one systemic frustration you experience regularly in your practice. What is one bias or assumption you hold about *why* this problem hasn't been fixed yet? Our goal here is to surface our viewpoints so we can learn to reframe them."

Slide 3: The Lineage of Physician Innovators

- **Visual Format:** A split-screen or four-quadrant profile slide highlighting diverse innovation types (Process, Device, System, Organizational).
- **Narration (Speaker Notes):** "When we talk about designing the system, we aren't speaking abstractly. Physicians have always been at the vanguard of innovation. Consider **Dr. Thomas Fogarty**, who invented the balloon embolectomy catheter, fundamentally changing vascular surgery (Device Innovation). Look at **Dr. Peter Pronovost**, whose creation of the central line checklist saved thousands of lives and revolutionized patient safety protocols globally (Process Innovation). Consider **Dr. Patricia Bath**, who invented the Laserphaco Probe for cataract surgery (Technological Innovation). Or look at organizational leaders who transitioned systems from reactive 'sick care' to proactive 'health assurance' models, such as the transformative growth seen right here at Jefferson Health over the last decade (Organizational Innovation). You are stepping into this exact lineage."

Slide 4: The Innovator's "Hero's Journey"

- **Visual Format:** A visual map or roadmap showing the three-course arc. Center the learner ("YOU") in the journey.
- **Narration:** "This certificate is designed to unlock a skill set you already possess as clinicians. We will treat this program as a personal and professional quest—a Hero's Journey. Course 1, which you are starting now, is 'Sparking Innovation: From Insight to Opportunity,' where you will learn to see your daily world differently. Course 2 is 'Building the Solution: Design, Plan, and Pitch'. Finally, Course 3 is 'Driving Impact: From Project to Lasting Change,' where you bring your vision to life. By the end of this journey, you won't just have new knowledge; you will have a tangible playbook and the foundation for an executive one-pager to make your ideas a reality."

Slide 5: Expectations & Mental Frameworks

- **Visual Format:**
- **Narration:** "To succeed, you must cultivate two foundational elements: a **Growth Mindset** (coined by Carol Dweck) and **Psychological Safety** (defined by Amy Edmondson). As you progress, you will be asked to share half-baked ideas and expose systemic flaws. We must collectively commit to maintaining psychological safety in our discussion boards. If we cannot be vulnerable about what is broken, we cannot design what is next."

Multimedia Integration: The Freakonomics Podcast

- **Format:** Embedded audio player.
- **Context Text:** "Listen to *Freakonomics: How to Succeed at Failing (Part 2)*, which examines the Morbidity and Mortality (M&M) conference through the lens of psychological safety."
- **Journal Prompt:** "As you listen, reflect on your own department's case review conferences. Are they conducted to assign blame, or to uncover systemic design flaws? In your private course journal, write down one way your department currently penalizes 'intelligent failure'."

Part 1 Summary Slide

- **Visual:** Bulleted recap with a forward-pointing arrow.
- **Summary Text:**
    - **The Gap:** Traditional training creates clinical experts, but not necessarily system innovators.
    - **The Mission:** Move from an employee *within* the system to an innovation leader *shaping* the system.
    - **The Journey:** A three-part progression from identifying opportunities to driving lasting impact.
    - **The Mindset:** Success requires agility, first principles thinking, and embracing intelligent failure

Chapter 1 Knowledge Check: Setting the Stage (5 Questions)

Q1: The Innovation Mindset (Remembering)

- **Prompt:** How does this module define the "innovation mindset" for a healthcare professional?

- **Correct Answer:** The internal conviction that you are a designer of the healthcare system, empowered to bridge the gap between current limitations and future possibilities, rather than just an employee within it.

Q2: Physician Innovators (Understanding)

- **Prompt:** Dr. Peter Pronovost's implementation of the central line checklist is primarily an example of which type of innovation?
- **Correct Answer:** Process Innovation.

Q3: Mental Frameworks (Applying)

- **Prompt:** A clinical team experiences an unexpected spike in patient readmissions after implementing a new discharge protocol. Instead of assigning blame, the attending physician says, "This is valuable data showing us where our protocol is fragile." This response best demonstrates which mental framework?
- **Correct Answer:** Embracing Intelligent Failure.

Q4: Psychological Safety (Evaluating)

- **Prompt:** According to Amy Edmondson, which of the following is the best indicator that a clinical department has high psychological safety?
- **Correct Answer:** Junior residents frequently point out potential workflow errors to attending physicians without fear of interpersonal penalty.

Q5: The Hero's Journey (Understanding)

- **Prompt:** In the context of the Innovator's "Hero's Journey" outlined in this certificate, what is the primary goal of Course 1?
- **Correct Answer:** Sparking innovation by learning to see the daily clinical world differently and moving from insight to a rigorously vetted opportunity.

Part 2: The Macro Trends & The Case for Change (Chapters 2 & 3)

Multimedia Integration: Video

- **Format:** Embed YouTube Video (https://www.youtube.com/watch?v=GjRAHUHlOdo).
- **Context Text:** "Before we look at the specific data driving healthcare's transformation, watch this brief breakdown of healthcare supply and demand. As

you watch, consider: How do these macroeconomic principles manifest in your daily clinical workflows?”

Slide 1: The Economic & Structural Reality

- **Visual Format:** Dual-Axis Data Visualization (Clean, high-contrast chart).
- **Prompt to Generate Image:** > *A sleek, corporate data visualization slide on a dark navy background. The left side features an upward trending line graph labeled 'National Health Expenditures reaching $5.2 Trillion'. The right side features two comparing bar charts showing 'Independent Practice' shrinking and 'Hospital Employment' growing. Clean sans-serif typography, highly legible, modern aesthetic.*
- **Narration (Speaker Notes):** "To understand our current environment, we must first look at the structural forces at play. U.S. healthcare spending continues to rise exponentially, now accounting for nearly one-fifth of the nation's economy and reaching $5.2 trillion in 2024. Alongside this financial pressure is a massive structural shift in how we work. Between 2012 and 2022, independent private practice fell drastically, while the share of physicians employed by large health systems surged. We are increasingly working within massive, complex organizations, which fundamentally changes our autonomy and decision-making authority. To effect change now, you cannot just optimize your own clinic; you must know how to navigate and influence a large enterprise."

• **References:** CMS NHE Fact Sheet; American Medical Association Practice Ownership Report.

Slide 2: The Human Cost: System Waste & Burnout

- **Visual Format:** Infographic with two stark, contrasting statistics.
- **Narration (Speaker Notes):** "These massive, complex systems are currently incredibly inefficient. Roughly 25% of total U.S. healthcare spending goes toward administrative costs and waste, not patient care. But the true cost of this inefficiency is human. Over half of physicians report symptoms of burnout. Crucially, the top driver of this burnout is not patient volume; it is bureaucratic tasks and administrative bloat. This is a profound insight. The burnout epidemic is not a personal failing of resilience; it is a system design problem. The workarounds we invent daily to survive a poorly designed EHR or a broken admission process are the exact friction points that are ripe for innovation."

References: Shrank WH et al., JAMA (2019); Medscape Physician Burnout Report (2024).

Required Reading: Article

- **Format:** Embedded hyperlink. https://medium.com/@TylerBeauchamp/healthcare-s-trillion-dollar-problem-90438f4164dc
- **Context Text:** "We just discussed the human cost of a poorly designed system. To dive deeper into the mechanics of this waste, read Tyler Beauchamp's breakdown of 'Healthcare's Trillion-Dollar Problem'. Pay special attention to the diagram categorizing the domains of care failure."

Slide 3: The Knowledge Explosion & New Technologies

- **Visual Format:** Conceptual Animation or Timeline Diagram.
- **Narration (Speaker Notes):** "At the same time, the tools we use are evolving faster than our ability to adapt. In 1950, it took 50 years for medical knowledge to double; today, that doubling time is projected at just 73 days. We can no longer rely on static training. Furthermore, 66% of physicians are now using some form of AI in practice, yet formal training on how to oversee these tools safely remains severely limited. But the upside is tremendous. We see digital health innovations, like the SMART remote monitoring program, significantly reducing ER visits for COPD and heart failure by shifting care from reactive to proactive. We also see virtual reality transforming procedural training by allowing learners to rehearse complex scenarios without patient risk. Technology will reshape care, but it requires clinical innovators to ensure it does so safely and effectively."

References: Densen P. (2011); Isaranuwatchai W et al., JMIR (2018); AMA AI Report.

Slide 4: The Consumer Shift & The Digital Front Door

- **Visual Format:** Infographic showing a patient journey (Smartphone to Provider).
- **Narration (Speaker Notes):** "Patients are no longer just passive recipients of care; they are consumers who expect the same transparency, digital access, and convenience they experience in every other industry. This requires organizations to rethink access through infrastructure like the 'Digital Front Door.' A true digital front door doesn't just offer a basic patient portal; it creates a frictionless journey. For example, Jefferson Health has moved beyond basic telehealth to create a comprehensive digital ecosystem, leveraging AI-driven triage to allow patients to navigate their care journey seamlessly. In a consumer-driven environment, intentional system design is what separates a frustrating patient experience from a world-class one."

Slide 5: The Leadership Gap

- **Visual Format:** Bar chart highlighting the gap between "Interest in Leadership" and "Formal Training."
- **Narration (Speaker Notes):** "This brings us to the ultimate gap, and the reason you are here. The data shows that 67% of practicing physicians want to pursue leadership roles to have a voice in organizational decisions. Yet, only 18% receive any executive or business training, and very few organizations offer formal leadership development. The COVID-19 pandemic clearly demonstrated that management and systems-thinking skills, often viewed as 'soft,' are actually critical to health and well-being. Academic Health Centers face this acutely—they have the clinical expertise but often lack the cultural incentives and interdisciplinary integration to translate ideas into scalable improvements. We need individuals who can bridge clinical practice with business and technology strategy."

References: Jackson Physician Search (2023); Rotenstein LS et al., Academic Medicine (2021).

Part 2 Summary & Knowledge Check

- **Visual Format:** Bulleted recap with a bold, concluding statement.
- **Summary Text:**
    - **System Strain:** Rising costs, administrative waste, and consolidation are fundamentally changing practice models.
    - **Human Toll:** Bureaucratic friction is the primary driver of the burnout epidemic.
    - **Pace of Change:** Exponential growth in medical knowledge and AI outpaces traditional training.
    - **The Opportunity:** Incremental adaptation is no longer sufficient; the industry requires guided, system-aware innovation led by frontline clinicians.
- **Narration (Speaker Notes):** "To summarize our case for change: Healthcare is shifting toward a system that is data-driven, consumer-responsive, and operationally incredibly complex. Incremental adaptation is no longer sufficient. Without deliberate leadership, technological advancement will not translate into improved outcomes. You are uniquely positioned to lead this change because you already navigate uncertainty and solve complex problems in real-time. It is time to translate your clinical expertise into tools for shaping the future of care."

- **Format:** 2-Question Interactive Block.
- *Question 1:* Drag and drop the primary driver of physician burnout to the top of the list (Answer: Bureaucratic tasks/EHR Charting).
- *Question 2:* True/False - The shift toward healthcare consumerism means organizations only need to focus on improving clinical quality. (Answer: False).

Chapter 2 Knowledge Check: The Macro Trends (5 Questions)

Q1: Structural Shifts (Remembering)

- **Prompt:** Between 2012 and 2022, which major structural shift occurred regarding physician practice models?
- **Correct Answer:** A drastic decrease in independent private practice and a surge in physicians employed by large health systems.

Q2: Healthcare Economics (Understanding)

- **Prompt:** Approximately what percentage of total U.S. healthcare spending is attributed to administrative costs and waste?
- **Correct Answer:** 25%

Q3: Burnout Epidemic (Analyzing)

- **Prompt:** While patient volume and acuity are high, the data shows that the *primary* driver of the physician burnout epidemic is:
- **Correct Answer:** Bureaucratic tasks and administrative bloat (e.g., EHR charting).

Q4: Supply and Demand (Evaluating)

- **Prompt:** Based on the "Healthcare Supply and Demand" concepts, why does simply hiring more physicians often fail to solve access issues in a poorly designed system?
- **Correct Answer:** Because it adds expensive supply without addressing the underlying systemic inefficiencies or administrative bottlenecks that limit throughput.

Q5: The Modern Dilemma (Applying)

- **Prompt:** You are an attending physician managing a complex workflow. You notice a recurring delay in patient transport. According to the "Modern Clinician's Dilemma," what is the most likely barrier preventing you from solving this?

- **Correct Answer:** The absence of a structured innovation process and the organizational authority/safety to redesign the workflow.

Chapter 3 Knowledge Check: The Case for Change (5 Questions)

Q1: The Knowledge Explosion (Remembering)

- **Prompt:** In 1950, medical knowledge doubled every 50 years. What is the currently projected doubling time?
- **Correct Answer:** 73 days.

Q2: Technological Acceleration (Understanding)

- **Prompt:** While 66% of physicians report using some form of AI in practice, what is the critical gap identified in the macro trends?
- **Correct Answer:** Formal training on how to oversee, evaluate, and safely implement these tools remains severely limited.

Q3: Healthcare Consumerism (Applying)

- **Prompt:** A hospital implements a "Digital Front Door" strategy. Which of the following initiatives best represents this concept?
- **Correct Answer:** An AI-driven, mobile-friendly triage system that allows patients to seamlessly schedule appointments and communicate with their care team before entering the physical clinic.

Q4: The Leadership Gap (Analyzing)

- **Prompt:** The data shows 67% of physicians want to pursue leadership roles, yet only 18% receive formal business training. What is the most significant consequence of this gap for Academic Health Centers?
- **Correct Answer:** They possess deep clinical expertise but lack the interdisciplinary management skills required to translate clinical ideas into scalable, system-wide improvements.

Q5: The Trillion-Dollar Problem (Evaluating)

- **Prompt:** Referring to Tyler Beauchamp's article, which category of care failure requires the most urgent intrapreneurial intervention to reduce macro-level waste?
- **Correct Answer:** Administrative complexity and failures of care coordination.

Part 3: The Opportunity & The Clinician as an Innovator (Chapters 4 & 5)

Slide 1: Physicianship = Innovation Leadership

- **Visual Format:** Overlapping Venn Diagram or Trait Matrix.
- **Prompt to Generate Image:** > *A clean, modern Venn diagram graphic on a dark background. The left circle is titled 'Physicianship' and the right circle is 'Innovation Leadership'. The overlapping center text should include: Empathy, Curiosity, Analytical Prowess, Resiliency, Comfort with Ambiguity. Use a professional healthcare color palette of navy blue and bright teal.*
- **Narration (Speaker Notes):** "When physicians think of 'innovators,' they often picture tech founders in Silicon Valley. But let's look at the core traits required for both physicianship and innovation. Both require profound empathy to understand the user or patient's pain. Both demand analytical prowess to sift through noise and find the signal. Both require comfort with ambiguity and the resiliency to pivot when a treatment plan—or a project—fails. You do not need to learn an entirely new personality to be an innovator; you simply need to apply your existing clinical traits—empathy, analytical prowess, and comfort with ambiguity—to operational and systematic problems.

Slide 2: The Bridge: Diagnosis = Design Thinking

- **Visual Format:** Side-by-side Flowchart.
- **Prompt to Generate Image:** > *A side-by-side process flow diagram comparing two pathways. Pathway 1 (Left, Blue): 'The Clinical Method' showing steps: History → Physical/Labs → Diagnosis → Treatment Plan. Pathway 2 (Right, Green): 'Design Thinking' showing steps: Empathize → Define → Ideate → Prototype/Test. Draw visual connecting lines between History/Empathize, Diagnosis/Define, and Treatment/Prototype.*
- **Narration (Speaker Notes):** "Let's make this tangible. The most common framework for innovation is 'Design Thinking.' At first glance, it sounds like business jargon. But map it to your daily clinical workflow. When you take a patient's history, you are performing the first step of Design Thinking: **Empathize**. When you synthesize the physical exam and labs into a diagnosis, you are performing the second step: **Define** the problem. When you formulate a differential and select a treatment plan, you are **Ideating** and **Prototyping** a solution. You run this innovation cycle twenty times a day. We are just changing the 'patient' from a human being to a broken workflow or system."

Integrating "The Intrapreneurship Formula" by Sandra Lam

- **Canvas/CourseArc Format:** "Required Reading" block before Slide 3 in this section.
- **Context Text:** *"Read Chapter 4 of 'The Intrapreneurship Formula' by Sandra Lam. As you read, focus on how Lam breaks down intrapreneurship from a vague concept into a structured, replicable formula."*
- **Visual: Include the formula:**
    - Use the formula and the venn diagram from the intrapreneurship formula book (page 48)
- **Updated Narration for Intrapreneurship Slide:** "As Lam outlines in Chapter 4, intrapreneurship is not just a personality trait; it is a calculated approach to driving change from within. The Intrapreneurship Formula requires you to balance your clinical vision with the realities of your organizational environment. You are leveraging your hospital's existing resources, brand, and patient population to scale your solution, which means you must master the art of internal stakeholder alignment—a skill we will heavily focus on in our Capstone."

Slide 3: Intrapreneurship vs. Entrepreneurship

- **Visual Format:** Split-screen definitions with iconography.

Reference: What Is Intrapreneurship and Why It Matters in 2026 - Hustle Inspires Hustle ™

- **Narration (Speaker Notes):** "Now that we know *how* you innovate, *where* do you do it? We often conflate innovation with Entrepreneurship—leaving the hospital to start a new company from scratch. But the most immediate impact you can make is through **Intrapreneurship**. An intrapreneur utilizes entrepreneurial principles to create, develop, and implement innovative solutions *within* an existing organization. It is navigating the constraints, politics, and resources of your current health system to improve it from the inside out. It requires a different type of political savvy, but it carries immense potential for scale. As Lam outlines, the formula requires balancing your clinical vision with your organizational environment, leveraging existing resources and mastering stakeholder alignment."

Slide 4: Idea vs. Opportunity

- **Visual Format:** Add the image from powerpoint slide #23 from HII Source PPT above

- **Narration (Speaker Notes):** "As an intrapreneur, your first major hurdle is understanding the difference between an Idea and an Opportunity. An idea is simply a thought or a suggestion. 'We should use AI to write notes' is an idea. On its own, its value is zero. An **Opportunity** is an idea that has been rigorously vetted for feasibility, viability, and desirability within your specific organization. It has a clear path to execution. In this certificate program, our goal is to stop generating ideas, and start engineering actionable opportunities."

Discussion Post 2: The Worked Example (Ungraded)

- **Format:** Ungraded, embedded example in Canvas showing a "Good vs. Bad" problem formulation.
- **Context Text:** *"Moving from an 'Idea' to an 'Opportunity' requires rigorous problem framing. Let's look at a worked example of how to elevate your thinking from a superficial complaint to a root-cause analysis. Review the scenario below to prepare for your final graded discussion post at the end of this module."*
- **The Scenario Content:**
    - **The "Bad" Idea (Superficial):** *"My idea is to buy iPads for all the nurses so they can do discharge teaching faster."*
    - **The "Productive Struggle" Breakdown (Audio/Video Narration or Text):** "Why is this a bad start? It jumps straight to a solution (iPads) without defining the actual problem. It assumes technology fixes human workflows. Let's run a root-cause analysis."
    - **The "Good" Reframing (The Opportunity):** *"Chief Complaint: Discharge times are delayed by an average of 3 hours on the med-surg floor. Root Cause: Nurses do not have a standardized, easily accessible checklist for patient education, causing redundant trips to the nursing station. The Opportunity: To design a standardized, mobile-friendly discharge workflow that reduces nursing administrative time and speeds up bed turnover."*

Part 3 Summary Slide

- **Visual Format:** Bulleted recap.
- **Summary Text:**
    - **You are already equipped:** Clinical skills (empathy, diagnosis) map directly to innovation frameworks (Design Thinking).

- **Your Arena:** Intrapreneurship is the act of driving entrepreneurial change from *within* your existing health system.
- **Your Goal:** Move past generating "Ideas" (value = zero) to rigorously defining actionable "Opportunities."

Chapter 4 Knowledge Check: The Clinician as an Innovator (5 Questions)

Q1: Physicianship vs. Innovation (Understanding)

- **Prompt:** Which of the following core clinical traits directly overlaps with the skills required for innovation leadership?
- **Correct Answer:** Comfort with ambiguity and analytical prowess.

Q2: Diagnosis = Design Thinking (Applying)

- **Prompt:** When a physician takes a detailed History of Present Illness (HPI) from a patient to understand their daily struggles, they are performing the equivalent of which Design Thinking phase?
- **Correct Answer:** Empathize.

Q3: Bridging the Frameworks (Analyzing)

- **Prompt:** Formulating a differential diagnosis and selecting a targeted treatment plan maps most closely to which steps in the Design Thinking process?
- **Correct Answer:** Ideate and Prototype.

Q4: The Empathy Gap (Evaluating)

- **Prompt:** A committee designs a new rounding schedule without consulting the night-shift nurses. The initiative fails within a week. Which phase of Design Thinking was critically bypassed?
- **Correct Answer:** Empathize/Define.

Q5: First Principles Thinking (Remembering)

- **Prompt:** In the context of clinical innovation, what does "First Principles Thinking" require you to do?
- **Correct Answer:** Break down complex systemic problems to their most fundamental truths rather than relying on convention or "how we've always done it."

Chapter 5 Knowledge Check: Intrapreneurship vs. Entrepreneurship (5 Questions)

Q1: Defining Intrapreneurship (Remembering)

- **Prompt:** According to Sandra Lam's *Intrapreneurship Formula*, what defines an intrapreneur?
- **Correct Answer:** Someone who utilizes entrepreneurial principles to create and implement innovative solutions from *within* an existing organization.

Q2: The Formula Components (Understanding)

- **Prompt:** Lam's Intrapreneurship Formula requires the intersection of three specific elements. What are they?
- **Correct Answer:** Individual Mindset, Learned Skillsets, and the Organizational Environment.

Q3: Idea vs. Opportunity (Applying)

- **Prompt:** A colleague says, "We should use virtual reality headsets to distract pediatric patients during blood draws." In the vocabulary of this course, this statement is currently an:
- **Correct Answer:** Idea (Value = Zero).

Q4: Assessing Viability (Analyzing)

- **Prompt:** To transform the "Virtual Reality" idea from the previous question into a true "Opportunity" (Value > Zero), what must the intrapreneur do?
- **Correct Answer:** Rigorously vet the idea for feasibility, viability, and desirability within their specific hospital's constraints.

Q5: The Intrapreneur's Advantage (Evaluating)

- **Prompt:** Compared to a startup entrepreneur, what is the primary strategic advantage an intrapreneur possesses?
- **Correct Answer:** Direct access to an existing patient population, established brand trust, and internal organizational resources.

Part 4: Assessments & The Bridge to Module 2

Discussion Post 3: Graded Peer Evaluation (The "M&M" of Innovation)

- **Format:** Graded, mandatory asynchronous discussion board in Canvas.

- **Context Text:** *"This is the capstone exercise for Module 1. We are going to apply the rigor of a clinical Morbidity and Mortality (M&M) conference to operational failures. You will propose a reframed problem, and your peers will act as your diagnostic sounding board."*
- **The Prompt:**
    1. **The Initial Presentation:** Select one recurring operational, clinical, or administrative failure from your own Opportunity Radar.
        - Provide a 1-sentence "Chief Complaint" (the superficial problem).
        - Perform a brief "5 Whys" root-cause analysis.
        - Present your "Reframed Opportunity" (How you have shifted this from a complaint to an actionable target).
    2. **The Peer "Differential" (Reply Requirement):** You must formally review at least two peers' presentations. In your review, you must identify one "Blind Spot" in their root-cause analysis. Provide a constructive "Differential Diagnosis" for why their proposed opportunity might face organizational resistance (referencing Sandra Lam's *Intrapreneurship Formula* or the concepts of psychological safety we discussed earlier).

Comprehensive Module Knowledge Check

- **Format:** Graded Quiz.

1. The True Source of Burnout The data presented in the macro trends indicates that the sharp rise in physician burnout is fundamentally a problem of: A) Insufficient personal resilience and mindfulness training. B) System design, driven primarily by bureaucratic tasks and administrative bloat. *(Correct)* C) Increasing patient acuity and the aging population. D) The shift from private practice to hospital employment.

2. The Intrapreneurial Environment According to Sandra Lam's *Intrapreneurship Formula*, if a physician has a strong innovation mindset and excellent design-thinking skills, but operates in a hospital department that strictly punishes deviations from standard protocol, what is the most likely outcome? A) The physician will successfully launch a startup. B) Innovation will stall due to a lack of psychological safety in the Organizational Environment. *(Correct)* C) The physician's skillset will override the departmental culture. D) The department will naturally adapt to the physician's mindset.

3. Applying Design Thinking A surgical team wants to reduce OR turnover times. Before buying new sterilization equipment, they spend three days shadowing the environmental

services (EVS) staff to understand their physical workflow constraints. Which Design Thinking phase does this represent? A) Empathize *(Correct)* B) Define C) Prototype D) Test

4. Intelligent Failure vs. Preventable Error Drawing on Amy Edmondson's concepts (as discussed in the Freakonomics podcast), which of the following represents an "Intelligent Failure"? A) A resident skips a mandatory checklist step due to fatigue, resulting in a near-miss medication error. B) A department pilots a novel, small-scale AI scribing tool; the tool struggles with complex medical jargon, so the pilot is paused to retrain the algorithm. *(Correct)* C) A hospital ignores known software bugs in their EHR, leading to lost billing revenue. D) A surgeon tries a highly experimental technique on a complex case without consulting the literature.

5. The Digital Front Door Strategy In the context of the shift toward healthcare consumerism, a robust "Digital Front Door" is best defined as: A) Upgrading the hospital's public-facing marketing website. B) Creating a frictionless, proactive digital ecosystem for patient access, triage, and communication. *(Correct)* C) Replacing physical front desk staff with automated kiosks. D) Mandating that all initial consults be conducted via telehealth.

6. Idea vs. Opportunity You write down: "We should use smartwatches to monitor post-op cardiac patients at home." At this exact moment, this statement is best classified as: A) An Opportunity B) A Value Proposition C) An Idea (Value = Zero) *(Correct)* D) A Prototype

7. First Principles in Action To solve chronic overcrowding in the ED waiting room, a hospital administrator suggests building a larger waiting room. A physician utilizing "First Principles Thinking" would most likely respond by asking: A) "How much will the new construction cost?" B) "Why are patients waiting at all, and what are the fundamental bottlenecks preventing immediate rooming?" *(Correct)* C) "What are other hospitals in our region doing to expand their waiting areas?" D) "Can we hire more triage nurses for the larger space?"

8. The Clinical Bridge Which step of the clinical diagnostic process is most analogous to the "Define" phase of Design Thinking? A) Taking the History of Present Illness (HPI). B) Ordering a metabolic panel. C) Synthesizing the data into a specific formal diagnosis. *(Correct)* D) Prescribing a targeted antibiotic.

9. The Modern Clinician's Dilemma What is the core paradox of the "Modern Clinician's Dilemma" as introduced in Chapter 1? A) Physicians are trained to be experts in technological systems but lack clinical skills. B) Physicians are expert problem-solvers in patient care, but feel disempowered and lack the frameworks to solve the systemic issues surrounding that care. *(Correct)* C) Healthcare systems have too many financial resources

but not enough ideas. D) Patients demand consumer-level convenience, but technology is not advanced enough to provide it.

10. Reframing the Problem Which of the following represents the most rigorous, actionable problem statement suitable for the "Opportunity Radar"? A) "The EHR is terrible and takes too much time." B) "Residents are unhappy with the feedback process." C) "The manual, paper-based consent process for routine endoscopies causes an average 15-minute delay per case and redundant work for the nursing staff." *(Correct)* D) "We need to buy iPads to fix the discharge delays."

11. The Leadership Gap Constraint Despite 67% of physicians expressing a desire for leadership roles, what structural barrier frequently prevents Academic Health Centers from achieving systemic innovation? A) A lack of clinical knowledge among the physician workforce. B) A culture that highly incentivizes interdisciplinary collaboration. C) The fact that only 18% of physicians receive formal executive or business training. *(Correct)* D) An overabundance of administrative staff.

12. The Power of Intrapreneurship Why might an intrapreneurial physician at a large academic medical center have a higher likelihood of initial scale than a solo entrepreneur launching a healthcare app? A) Intrapreneurs face less regulatory scrutiny. B) Intrapreneurs do not have to worry about value propositions. C) Intrapreneurs can leverage the established trust, patient population, and infrastructure of their existing health system. *(Correct)* D) Intrapreneurs are immune to organizational politics.

13. Growth Mindset and Innovation According to Carol Dweck's framework, how does a physician with a "Growth Mindset" view a rejected pitch for a new quality improvement initiative? A) As proof that the organization is fundamentally broken and cannot be fixed. B) As a reflection of their own lack of innate leadership ability. C) As valuable feedback highlighting gaps in their stakeholder alignment or data presentation. *(Correct)* D) As a personal attack by the executive committee.

14. The 73-Day Problem The statistic that medical knowledge now doubles every 73 days primarily argues for which of the following? A) Extending the length of medical school and residency programs. B) Shifting medical education away from rote memorization toward systems-thinking, agility, and continuous technology integration. *(Correct)* C) Reducing the number of complex procedures performed. D) Relying exclusively on AI to make diagnostic decisions.

15. The Final Objective By the conclusion of Course 1 (The Capstone), what is the ultimate, tangible output expected from the frameworks introduced in this module? A) A peer-reviewed medical journal publication. B) A fully coded and deployed software

application. C) A completed executive one-pager that synthesizes a reframed problem, a value proposition, and a targeted stakeholder pitch. *(Correct)* D) A certification in Lean Six Sigma.

Innovation Leader Playbook Entry #1

- **Format:** The first entry in the learner's personal "Innovation Leader Playbook": a living document you can look at throughout the program. *(Recommendation: Design this as a fillable, branded PDF or a locked Notion/Google Doc template that they copy. This builds a sense of prestige and continuity)*.
- **Please include your reasoning and thought-process for every step**
    - **Task 1 (Evaluate): My Innovator's Starting Point**
        - **Prompt:** "Critically evaluate your own mindset, motivations, and work environment to establish a baseline for your innovation journey."
            - **Goals & Fears:** What is your primary goal for taking this certificate program? What is one fear or apprehension you have about this process?
            - **Optimizing Your Experience:** How can you proactively structure your time or mindset to get the most out of this program?
            - **Personal SWOT Analysis:** Complete a personal SWOT analysis (Strengths, Weaknesses, Opportunities, Threats) related to your capacity for innovation. In your 'Weaknesses' section, identify one aspect of a 'Fixed Mindset' you sometimes exhibit. In your 'Opportunities' section, describe the current level of psychological safety on your primary team and identify one opportunity to improve it.
            - Download Here (and link to the below file)

Personal_Innovation_SWOT_Worksheet.docx

Instructional Reasoning: This task perfectly operationalizes Dweck's Growth Mindset and Edmondson's Psychological Safety. By forcing the learner to explicitly name their "fears" and "fixed mindset" traits upfront, you are disarming the traditional medical ego and establishing a fertile ground for vulnerability and growth. Tying the "Threats" and "Opportunities" directly to their team's psychological safety connects personal introspection to systemic awareness.

- **Task 2 (Create): My Initial Idea Portfolio**
    - **Prompt:** "Source the initial seeds of innovation from your own experience. Briefly describe three distinct ideas, one from each of the following categories:"
        - **The Future Idea:** Describe a problem you'd like to solve or an idea you'd like to work on in the future.
        - **The Past Idea:** Describe a problem or project you've worked on in the past. What went well? What would you do differently now? How did a lack of stakeholder alignment or an unclear 'Value Proposition' contribute to the outcome?
        - **The Observed Idea:** Describe an interesting project or innovation you've seen a colleague or another department work on. What did you find compelling about it?

Instructional Reasoning: This utilizes the principle of "Scaffolded Recall." It proves to the physician that they are not starting from zero. By categorizing into Future, Past, and Observed, you prevent the common "blank page paralysis" that plagues highly analytical learners when asked to be creative.

- **Task 3 (Create): My Opportunity Radar**
    - **Prompt:** "Establish your 'Opportunity Radar,' which will be the foundation for your work throughout this course. Observe your daily work environment and identify **5 problems, frictions, or opportunities for improvement**. These should be specific to your life as a physician. For each item, write a one-sentence problem statement. The goal is to generate a diverse list of potential innovation projects."
        - **Examples:**
            - "The process for consenting patients for routine procedures is manual, redundant, and consumes too much resident time."
            - "There is no standardized way for attending physicians to give residents meaningful feedback on their clinical documentation in the EHR."

    - "Patients frequently miss their post-op appointments because the scheduling and reminder system is confusing."
    - "Our department's grand rounds are poorly attended because the content is not perceived as relevant to daily practice."

Instructional Reasoning: This is the most critical step of the module. It translates the macro-level "Trillion Dollar Problem" into their micro-level daily reality. Generating five distinct statements forces volume, which is key in the "Ideate" phase of Design Thinking. They will inevitably return to this list in Course 1, Module 7 (The Capstone) to select the single most viable project to build their executive pitch around.

Innovation Leader Playbook: Master Grading Rubric

Grading Scale:

- **Exemplary (4 points):** Demonstrates mastery, deep synthesis of concepts, and executive-level readiness.
- **Proficient (3 points):** Demonstrates strong understanding and application, with minor gaps in depth or framing.
- **Developing (2 points):** Shows basic comprehension but relies on superficial analysis or incomplete application of frameworks.
- **Needs Improvement (1 point):** Fails to apply course concepts; responses are incomplete, highly defensive, or lack clinical/organizational relevance.

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1. Analytical Depth & Problem Framing (The "Diagnosis")

Evaluates the learner's ability to move from a surface-level "Chief Complaint" to a systemic root cause.

- **Exemplary (4):** Relentlessly interrogates the problem. Successfully isolates the systemic or structural root cause (using the "5 Whys" or similar logic) rather than blaming individuals or stopping at superficial symptoms. Problem statements are highly specific.
- **Proficient (3):** Identifies a valid problem and attempts a root-cause analysis, but the final problem statement may be slightly too broad or focus on a secondary symptom rather than the absolute root cause.

- **Developing (2):** Problem statements read as standard workplace complaints. Little evidence of digging beneath the surface. Jumps straight to proposing a solution before fully defining the problem.
- **Needs Improvement (1):** Problem statements are vague, purely emotional, or entirely misdiagnose the systemic issue.

2. Application of Innovation Frameworks (The "Methodology")

Evaluates how accurately the learner integrates module-specific tools (e.g., Design Thinking, Intrapreneurship Formula, Opportunity Radar).

- **Exemplary (4):** Seamlessly and accurately maps clinical experiences to the prescribed innovation frameworks. Uses course vocabulary (e.g., "Intelligent Failure," "Psychological Safety," "Opportunity vs. Idea") correctly and in context to justify their logic.
- **Proficient (3):** Accurately applies the frameworks to their entries, though the connection between the clinical scenario and the theoretical framework may occasionally feel forced or slightly disconnected.
- **Developing (2):** Mentions frameworks in passing but does not actively use them to shape their analysis. Misapplies core concepts (e.g., confusing an "Idea" with an "Opportunity").
- **Needs Improvement (1):** Completely ignores the prescribed frameworks, relying entirely on personal opinion or outdated management assumptions.

3. Organizational Feasibility & Actionability (The "Reality Check")

Evaluates the learner's situational awareness as an intrapreneur navigating a complex health system.

- **Exemplary (4):** Demonstrates acute awareness of their own sphere of influence and organizational constraints (budget, IT limitations, political dynamics). Proposed opportunities are highly actionable and realistic for an intrapreneur to pursue.
- **Proficient (3):** Proposes actionable opportunities, but may slightly underestimate the organizational resistance or resources required to implement them.
- **Developing (2):** Ideas are highly theoretical or require completely unrealistic resources (e.g., "We should just build a brand new hospital wing"). Lacks an understanding of the intrapreneurial constraint.

- **Needs Improvement (1):** Ideas are entirely disconnected from the reality of healthcare operations or fall completely outside the learner's potential sphere of influence.

4. Reflective Vulnerability & Mindset (The "Leader's Ego")

Evaluates the learner's capacity for self-critique, growth mindset, and psychological safety.

- **Exemplary (4):** Exhibits high reflective vulnerability. Willingly and objectively identifies personal blind spots, fixed mindset triggers, or departmental failures without defensiveness. Treats past failures purely as data for future improvement.
- **Proficient (3):** Shows a willingness to reflect and adapt, though may slightly hedge when discussing personal weaknesses or departmental failures.
- **Developing (2):** Reflection is guarded or superficial. Exhibits signs of a fixed mindset (e.g., blaming external factors exclusively for past project failures without acknowledging personal leadership gaps).
- **Needs Improvement (1):** Highly defensive or arrogant tone. Refuses to acknowledge personal blind spots or systemic flaws, viewing the current state as unchangeable.

5. Executive Communication (The "Pitch")

Evaluates the clarity, conciseness, and professionalism of the written entry.

- **Exemplary (4):** Writing is exceptionally crisp, concise, and punchy. Uses active voice and formatting (bullet points, bold text) to make the entry easily scannable for an executive stakeholder. Zero fluff.
- **Proficient (3):** Clear and professional, but may occasionally lapse into overly academic or unnecessarily dense medical jargon when simpler terms would suffice.
- **Developing (2):** Disorganized or rambling. Requires the reader to work hard to find the core argument or problem statement.
- **Needs Improvement (1):** Unprofessional tone, significant grammatical errors, or incoherent narrative structure.

Faculty Guide for Scoring:

- **18 - 20 Points (Honors / Pass):** The entry is portfolio-ready. The learner is successfully thinking like an executive intrapreneur.

- **14 - 17 Points (Pass):** The entry is solid but requires minor refinement. Faculty should provide targeted feedback on sharpening the problem statement or leaning harder into the frameworks.
- **Under 14 Points (Revise & Resubmit):** The learner is still thinking strictly like a clinical reactor rather than an innovator. Faculty should intervene, push them back to the "Define" phase, and require a resubmission using the "5 Whys" methodology.

Summary Slide & Bridge to Module 2

- **Visual Format:** Closing Video (Talking Head + Text overlay).
- **Video Storyboard:** * **0:00 - 0:15 (On-Camera):** You (Dr. Naik) are back on screen, speaking directly to the learner with an encouraging, forward-looking tone.
    - *On-Screen Text:* Module 1 Complete: The Mindset is Set.
    - **0:15 - 0:30 (Graphic Overlay):** A transitional graphic showing a bridge connecting "Module 1: Mindset" to "Module 2: Leadership Identity."
- **Narration (Speaker Notes):** "Congratulations on completing Module 1. We must ask a critical question: Now that you understand the mindset of an innovator, how do you begin your own personal journey?

You have mapped out the macro forces changing medicine, and you have filled your Opportunity Radar with vital targets. But an idea is only as good as the leader who champions it. What are the core characteristics you need to cultivate to succeed in a complex matrixed health system?

You are no longer just reacting to the system; you are preparing to redesign it. But an opportunity is only as viable as the leader championing it. Having the right idea is only 10% of the battle; the other 90% is having the political savvy, the communication skills, and the leadership identity to get others to follow you.

In Module 2, we will move from *Mindset* to *Identity*. You will begin to define who you are as a leader, how you communicate your vision, and how you can inspire others to join your cause. The clinical problem is waiting; it is time to build the leader who will solve it. See you in Module 2."